

LINGYU (ALVIN) XU

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EDUCATION

Worcester Polytechnic Institute, Worcester, MA

Jan 2024 - Dec 2025

M.S. in Information Technology - System Design & Artificial Intelligence

GPA: 3.82 / 4.0

- Coursework: LLM, Application Development, Machine Learning, Data Mining, Data Visualization, SQL, Database Systems

Pennsylvania State University, State College, PA

Aug 2015 - May 2020

B.S. in Computer Science

- Coursework: Data Structures, Algorithms, System Programming, Network Programming, Statistics

EXPERIENCE

Research Assistant

July 2025 - Current

Worcester Polytechnic Institute

Worcester, MA

- **TimePre (Accepted at TMLR)**: Developed an advanced ML framework for probabilistic forecasting with multi-hypothesis density estimation and a channel-based reversible normalization technique to mitigate distribution shifts and preserve long-horizon non-stationary patterns during de-normalization. Built Python data-processing, evaluation, and validation workflows across multiple time-series datasets to improve reliability under distribution shifts.
- **CV-Arena**: Developed the Pro-Edit dataset and a Resolution Baseline Agent to support scalable benchmark evaluation, introducing a secure benchmark and a feedback-loop agentic architecture for collecting, filtering, and organizing high-fidelity image data for downstream model evaluation.
- **Moral Auditing of LLM Agents (Position Paper)**: Proposed a framework for auditing RLHF-trained LLMs as non-neutral negotiation agents; designed a five-scenario bargaining and persuasion stress test revealing a cooperative-assistant prior that helps in low-conflict settings but weakens under asymmetric pressure.

Full-Stack Software Development Engineer

Jun 2021 - Dec 2023

Dell Technologies, Infrastructure Solutions Group

Shanghai, China

- **Back-End APIs**: Built automated Python/Go/Java REST APIs for Azure HCI and VxRail Day-1 provisioning - covering Windows domain joining, security hardening, cluster validation, network/storage configuration, and VMware/VxRail pre-validation - reducing manual configuration overhead and accelerating deployment timelines by 20%.
- **Data Validation & Storage**: Authored REST APIs and SQL logic to collect, transform, and validate network, storage, and system configuration data, and designed database tables for server clustering with secure storage of encrypted configurations and deployment metadata, contributing to a 10% faster deployment process.
- **Front-End**: Designed and implemented a React/Angular/TypeScript interface for Day-1 operations - dynamic forms, filters, dropdowns, validation flows, and data visualizations - to guide operators through server pre-configuration, reducing setup errors by 15% and achieving over 93% coverage via contract and unit testing.
- **CI/CD & Containerization**: Led a 4-person team to build a Jenkins CI/CD pipeline and containerized provisioning microservices with Docker/Kubernetes; integrated MongoDB and PostgreSQL APIs for configuration storage, validation, and retrieval, cutting service startup failures by 25% and manual testing effort by 30%.
- **Quality & Agile**: Wrote Pytest component tests ensuring over 80% API coverage; collaborated with QA and product teams across Agile delivery cycles, maintaining clear, reusable code for deployment automation and system integration.

Software Development Engineer Internship

Mar 2021 - May 2021

Dell Technologies

Shanghai, China

- Engineered Python/Java APIs to automate Day-1 setup for Dell VxRail HCI systems - VMware configuration pre-validation, VxM network setup, and PostgreSQL table management - reducing incompatible configuration errors by 15%.
- Built responsive Angular UI components (dynamic forms, data filters) for the VxRail Day-1 & Day-2 operations team to pre-configure VLAN, authentication, and backup settings, increasing configuration efficiency by 10%.

PROJECTS

AI / Machine Learning

Agentic Advising Assistant

Worcester, MA

- **Architecture:** Architected a production-grade agentic advising platform using FastAPI, LangChain, and Google Gemini 2.5 Pro, orchestrating an Intent Classification Agent and a ReAct Adviser Agent over a Hybrid RAG pipeline (SQLite + ChromaDB) to automate complex academic advising workflows.
- **LLM & RAG Engineering:** Built a multimodal PDF extraction pipeline via Gemini Vision API for unstructured transcripts; designed a DB-first hybrid retrieval layer with course-aware chunking, score boosting, and query-rewrite fallback, eliminating hallucinations on course/program lookups.
- **Productionization:** Optimized the ReAct loop with 7 specialized tools, real-time SSE streaming, and Redis session memory (TTL + LRU); containerized with Docker Compose, achieving ~10s avg latency and a projected 60% reduction in manual advising workload.

Email Threat Classifier - Local LLM + Gmail Add-on

Worcester, MA

- **Model Post-Training:** Post-trained (QLoRA fine-tuned) Llama-3.2-3B-Instruct locally to classify incoming emails as spam, phishing/dangerous, or safe; ran 4-bit quantized on a single 12GB GPU for privacy-preserving, low-latency on-device inference without sending email content to third-party services.
- **API & Gmail Plugin:** Served the classifier behind a FastAPI REST endpoint and built a Gmail add-on that calls the API in real time to score, label, and warn on suspicious messages directly in the user's inbox.

American Sign Language Translation

Worcester, MA

- **Model Development:** Implemented and evaluated Dense Feedforward Networks, 2D Conv-RNNs, and DinoV2-enhanced networks to capture spatiotemporal features of sign videos, achieving 48.7% Top-1 accuracy and outperforming baseline models; demonstrated improved semantic matching via similarity-distribution visualizations.
- **Data Engineering:** Extracted 3D facial, pose, and hand landmarks from video using MediaPipe, then applied data cleansing and augmentation (rotation, flipping, masking) for robust downstream model training.

Intelligent Private Banking Portal - Capital One Capstone

State College, PA

- **Data Processing:** Built data-preparation workflows to aggregate transaction records, engineer spending features, resample imbalanced categories, and apply cross-validation for monthly net and category-specific spending prediction.
- **Analytics & Modeling:** Developed predictive models with TensorFlow, linear regression, and decision trees to forecast monthly spending patterns with 83% accuracy. Won prize at the 2019 Capstone Design Showcase.

Stock-Market Forecasting

Worcester, MA

- **ML Pipeline:** Applied PCA and polynomial feature expansion and cleaned missing data, then trained XGBoost and Random Forest models for price forecasting and trend classification, achieving $R^2 = 0.91$, MAE = 28.3, and 87% trend-classification accuracy.

Software Engineering

Blockchain Document & Payment Platform - 3rd Place, Harvard Hackathon

Boston, MA

- **Architecture:** Built a full-stack blockchain platform (Vue.js + Node.js) for decentralized medical-record sharing and XRP payment processing; backend handled document logic and blockchain integration on the Flare Coston2 testnet, reducing record-sharing evaluation time by 50%.
- **Recognition:** Awarded 3rd Place at the Harvard University Hackathon (EasyA x Flare Network x XRPL Commons), delivering an end-to-end demo of secure document exchange with on-chain settlement under a 36-hour build.

Android Banking App Development

Worcester, MA

- **Application:** Built a Java/JavaScript Android app simulating core banking operations - balance inquiry, transaction-history filtering, and dynamic sorting - backed by a structured SQLite database for persistent local storage.
- **Data Layer:** Engineered a normalized relational schema with interconnected tables; automated data generation with Python and implemented Java query logic to optimize retrieval performance and responsiveness.

SKILLS

Languages: Python, SQL, Go, Java, JavaScript, TypeScript, C, C++

Web & Backend: React, Angular, Node.js, FastAPI, REST APIs, GraphQL, HTML5, CSS

Data Engineering: ETL/ELT, Pandas, NumPy, PySpark, Apache Spark, Data Cleansing, Validation, Transformation

Databases & Cloud: PostgreSQL, MySQL, MongoDB, SQLite, ChromaDB, Redis, Azure, BigQuery, Google Cloud

ML & AI: PyTorch, TensorFlow, Scikit-Learn, Forecasting, Model Evaluation, RAG, LLM Applications

Tools & DevOps: Git, Docker, Kubernetes, Jenkins, Agile, Jira, Confluence